

Gender, Current Alcohol Use, and Sad or Hopeless Feelings among High School Students

113370209 王心妤
112310316 梁展育

research question

Main question

Is the proportion of sad or hopeless feelings different between female and male students?

Extension question

After considering current alcohol use, does the association between sex and sad/hopeless feelings still remain?

Data Collection · Variables

Data Collection

Source	YRBS 2007
Female	6,940
Male	6,893
Total	13,833

Variables

- X — Group variable**
WhatIsYourSex
Female / Male
- Y — Response variable**
SadOrHopeless
1 = Yes / 0 = No
- Z — Extension variable**
CurrentAlcoholUse
1 = Yes / 0 = No

Statistical Methods

MAIN ANALYSIS

- Two-proportion Z-test — test for significant difference
- Logistic regression

EXTENSION

- Adjusted predicted probability

Hypotheses

H_0 : $p_{\text{Female}} = p_{\text{Male}}$

No difference in proportions

H_1 : $p_{\text{Female}} \neq p_{\text{Male}}$

A difference exists (two-sided)

Statistical conclusion

Reject H_0

$$p < 0.001 \cdot z = 18.54$$

Type I / II error

Type I (α)	0.05
Type II (β)	Very low
Power	Near 1

Main Results — Two-Proportion Z-Test

37.2%

Female proportion
n = 6,940

22.7%

Male proportion
n = 6,893

18.54

Z-statistic

< 0.001

p-value
Reject H_0

+14.4 pp

Difference
Female – Male

Figures

Effect-Size Dashboard for Female vs Male Comparison

Risk difference

14.4 pp

Female proportion – Male proportion

Relative risk

1.64×

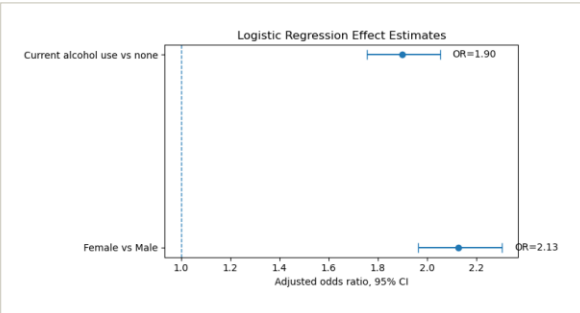
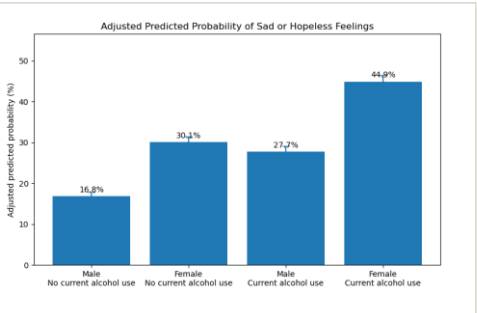
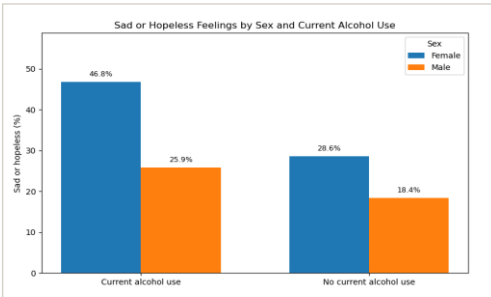
Female proportion / Male proportion

Odds ratio

2.01×

Female odds / Male odds

Two-proportion z-test: $z = 18.54$, $p < 0.001$; 95% CI for difference is saved in the summary table.



Key Results & Conclusion

① **Female consistently higher**

37.2% vs 22.7% (+14.4 pp)
Higher in both alcohol groups

② **Alcohol use → higher risk**

Both sexes show higher rates
with current alcohol use

③ **Both independently significant**

Adj. OR: Female 2.13×
Adj. OR: Alcohol 1.90×

④ **Highest-risk group**

Female + alcohol use: 44.9%
Highest predicted probability

Female students reported sad/hopeless feelings more often than male students. The association remains clear after considering current alcohol use. Observational survey data support association, not causation.

Limitations

- Observational survey data support association, not causation
- Responses were self-reported
- Some potentially important variables were not included
- The dataset was collected in 2007